

Presenting the simulator - 24 August

Written 2026-08-15. Nine days out.

Assumption, stated by Mridul: **you drive the demo and you do the talking**. No guided tour, no narration overlay, no onboarding for strangers. The site is a visual aid you operate live in front of a panel.

That changes the priorities completely. Everything in `SIMULATOR_REVIEW.md` about READMEs, licences, accessibility and tours is **backlog** - real, but not for the 24th. What follows is only what bites in a room with a projector.

The measured problem: nobody in that room can read this page

Font sizes declared in <code>index.html</code>	Count
<code><= 12px</code>	15 of 18
Most body text	10-11px

Body text is `--dim: #8d93b0` on `--panel: #181a2b` - about **5.3:1** contrast. Fine on your laptop, 40cm away, in a dark room.

A projector loses roughly half its contrast under normal room lighting, and the panel sits three to five metres back. Effective contrast lands near **2.5:1 at 10px**. They will read the big letter glyph and **nothing else** - not the part descriptions, not the decode panel, not the button labels.

This is the single highest-value fix for eval day and it is pure CSS.

The five things worth building

1. A Present mode - one toggle

Scale the whole UI $\sim 1.6\times$, raise `--dim` toward `--ink`, thicken panel borders, and drop the panels that are not being discussed. Bind it to a key so it survives a fumbled click.

Roughly one evening. Nothing else on this list comes close in value.

2. Preset camera buttons - **not* a tour*

The failure mode is you dragging the mouse mid-sentence hunting for the cam while the panel watches. That reads as unprepared, and a trackpad makes it worse.

Five buttons that jump instantly, no animation narration, no script:

[Overview] [Cam] [Linkage] [Electronics] [Dock]

You still say every word. The buttons only stop you fishing for an angle. The camera plumbing already exists - `__braillix.look()` was added for debugging and does exactly this.

2-3 hours.

3. Keyboard control

Reaching for five identical-looking buttons under pressure is a mistake waiting to happen.

Key	Action
Space	play / pause
->	step one cell
R	reset view
1-5	jump to preset view
X	X-ray

Also means you can drive it from a presenter remote if the lab has one. **30 minutes.**

4. A wires toggle

Right now Electronics shows boards **and** the loom together. The loom is visually busy, and anything that reads as clutter invites a question that eats your time.

Separate them so you can show boards first, then add the wiring when you choose to talk about it. **1 hour.**

5. Decide the shaft story now

5.2mm of motor shaft stands proud of the base plate. It is visible. Someone on that panel will point at it.

Two acceptable answers, and you must pick one before the day:

- **Present it.** *The simulator caught this - the shaft is 10mm, only 4.8mm fits in the cam

hub, so either the hub grows or the shaft gets cut.* This is the strongest possible answer: it proves the simulator does real work rather than being a pretty picture.

- **Hide it** with a flag in the code and do not raise it.

The failure mode is neither: being surprised by it and improvising. **Pick one this week.**

Zero-code items that matter more than any of the above

Run from localhost, not GitHub Pages

The payload is **3.7 MB**, of which 2.5 MB is one GLB. On lab wifi that is a blank screen with a spinner while a panel watches. From `run.bat` it is instant, and it works with the wifi down.

Have `run.bat` already running and the tab already open **before** you connect the projector.

Test at the projector's real resolution

Lab projectors are frequently **1024x768, 4:3**. The UI panels are fixed to the four corners and have only ever been checked at 16:9 and on a phone. At 4:3 they may well overlap.

Ten minutes with the actual projector removes the entire risk. If you cannot get the room, at minimum resize your browser window to 1024x768 and look.

Rehearse the exact click sequence

Write down the order and run it twice:

- 1 Braille 101 - Step through, point out the capital sign and the number sign
- 2 A1 - four cam indexes for two characters, the argument for multi-cell
- 3 CAB - Step, showing the cam reverse direction
- 4 X-Ray on - the linkages riding the cam
- 5 Electronics on - pod, driver, motor, the pogo interface
- 6 HELLO WORLD + Simulate - the whole thing running

Six moves. If you know them cold, nothing else on this page matters much.

Priority, honestly

	Item	Effort	If you only do...
1	Present mode (size + contrast)	1 evening	this one
2	Preset camera buttons	2-3 h	...and this
3	Rehearse the six-step sequence	1 h	...and this
4	Test at 1024x768	10 min	
5	Keyboard shortcuts	30 min	
6	Wires toggle	1 h	
7	Decide the shaft answer	0	

Items 1-3 are **one day total** and cover essentially all of the risk. Everything after that is polish.

Deliberately not before the 24th

- README, LICENSE, meta tags - **the panel is not browsing your repo**
- Accessibility and audio - genuinely important, entirely invisible on eval day
- Guided tour - you are the tour

- Draco compression - irrelevant when serving from localhost
- Multi-cell, hover labels, scale overlay - half-finished features read as broken

All of it stays in `SIMULATOR_REVIEW.md` for after.